

Mark Rober's CrunchLabs Acid vs. Lava Destruction Test

Time	Subtitle	Human Translation
0s	[Mark] Which has more destructive power? Lava or acid?	[Mark] Which has more destructive power? Lava or acid?
3s	To answer that question, they'll be going head-to-head	To answer that question, they'll be going head-to-head
6s	in a scientific face-off by testing each against seven objects,	in a scientific face-off by testing each against seven objects,
10s	including a turkey leg, an iPad, slabs of ice,	including a turkey leg, an iPad, slabs of ice,
14s	a bunch of toys, a baseball bat, a diamond,	a bunch of toys, a baseball bat, a diamond,
17s	and finally, an actual functioning car.	and finally, an actual functioning car.
20s	Oh, wow!	Oh, wow!
21s	To do this properly, I've invited YouTube's foremost expert	To do this properly, I've invited YouTube's foremost expert
24s	when it comes to chemical reactions...	when it comes to chemical reactions...
26s	It's good to go.	It's good to go.
27s	[Mark] ...with the buttery smooth voice of a high-school chemistry teacher...	[Mark] ...with the buttery smooth voice of a high-school chemistry teacher...
32s	[man 1] Oh no.	[man 1] Oh no.
33s	[Mark] ...NileRed.	[Mark] ...NileRed.
34s	And his opponent in this matchup will be YouTube's very own lava king.	And his opponent in this matchup will be YouTube's very own lava king.
39s	You love him much more than his neighbors do.	You love him much more than his neighbors do.
41s	-[man 2] The fire was almost to the top. - [neighbor] I just walked out here.	-[man 2] The fire was almost to the top. - [neighbor] I just walked out here.
45s	[man 2] It was cool.	[man 2] It was cool.
49s	[Mark] Kevin, The Backyard Scientist.	[Mark] Kevin, The Backyard Scientist.
51s	But there's a catch	But there's a catch
52s	'cause there's no way I'm just gonna watch.	'cause there's no way I'm just gonna watch.
55s	For each of the objects, I'll be testing a wild-card factor of destruction	For each of the objects, I'll be testing a wild-card factor of destruction
59s	that will be different for each object.	that will be different for each object.
1:01	So for the turkey leg, my wild-card factor of destruction	So for the turkey leg, my wild-card factor of destruction
1:04	may or may not be ravenous piranhas.	may or may not be ravenous piranhas.
1:07	Then for the baseball bat, it might involve Superman X-ray vision	Then for the baseball bat, it might involve Superman X-ray vision
1:11	that can do this in two seconds.	that can do this in two seconds.
1:13	That's pretty strong.	That's pretty strong.

1:15	And the nerd trash talk commenced immediately.	And the nerd trash talk commenced immediately.
1:18	-Nothing beats a volcano. -I think acid could.	-Nothing beats a volcano. -I think acid could.
1:20	Or the X factor. You don't know what I'm bringing.	Or the X factor. You don't know what I'm bringing.
1:23	And I do wanna mention, while I am no MrBeast,	And I do wanna mention, while I am no MrBeast,
1:26	winner of today's competition gets a car.	winner of today's competition gets a car.
1:28	-I'm not even tricking. -Really? You can drive it?	-I'm not even tricking. -Really? You can drive it?
1:31	Yeah. [laughs]	Yeah. [laughs]
1:32	-[Mark] But before we even got started... - Is this the right place?	-[Mark] But before we even got started... - Is this the right place?
1:35	[Mark] ...someone from our World's Largest Lemon Battery collab showed up.	[Mark] ...someone from our World's Largest Lemon Battery collab showed up.
1:40	Mr. William Osman.	Mr. William Osman.
1:42	But since he legitimately wasn't even invited,	But since he legitimately wasn't even invited,
1:44	I made him be referee.	I made him be referee.
1:46	So with the realization that I should have larger spare referee shorts on hand,	So with the realization that I should have larger spare referee shorts on hand,
1:51	it was time to commence our showdown of destruction,	it was time to commence our showdown of destruction,
1:54	starting first with Kevin's magma getting a chance to destroy a turkey leg.	starting first with Kevin's magma getting a chance to destroy a turkey leg.
1:58	Here we go. Turkey leg versus lava.	Here we go. Turkey leg versus lava.
2:01	[Mark] Oh yeah!	[Mark] Oh yeah!
2:02	Yeah! [laughs]	Yeah! [laughs]
2:07	Yes!	Yes!
2:09	-[Kevin] Burn! -[laughter]	-[Kevin] Burn! -[laughter]
2:11	-He's enjoying this way too much. -[Kevin] Boom!	-He's enjoying this way too much. -[Kevin] Boom!
2:17	More violent than I was expecting.	More violent than I was expecting.
2:18	Once it cooled down,	Once it cooled down,
2:20	we cracked open the crucible to inspect the damage.	we cracked open the crucible to inspect the damage.
2:23	[all] Oh!	[all] Oh!
2:24	[Mark] While it charred the outside, it turned the rest into a perfectly-cooked...	[Mark] While it charred the outside, it turned the rest into a perfectly-cooked...
2:28	It's fine.	It's fine.
2:29	[Mark] ...delicious Thanksgiving turkey leg,	[Mark] ...delicious Thanksgiving turkey leg,
2:32	which I know 'cause we definitely all gave it a taste.	which I know 'cause we definitely all gave it a taste.

2:35	-Yeah, that tastes... -[William] Come on!	-Yeah, that tastes... -[William] Come on!
2:37	[Mark] Which meant it was time for acid to enter the chat.	[Mark] Which meant it was time for acid to enter the chat.
2:43	In hindsight, this shouldn't have been surprising,	In hindsight, this shouldn't have been surprising,
2:45	considering what our own stomach acids do to food,	considering what our own stomach acids do to food,
2:48	but given a little bit of time, it was terrifyingly efficient.	but given a little bit of time, it was terrifyingly efficient.
2:52	-Guys, come here. What do you think? - [Mark] What's the clear and dark stuff?	-Guys, come here. What do you think? - [Mark] What's the clear and dark stuff?
2:56	The dark stuff is the turkey leg.	The dark stuff is the turkey leg.
3:00	How do we know you didn't throw it in the bush? Let me look.	How do we know you didn't throw it in the bush? Let me look.
3:03	[Mark] After a fruitless search for foul play,	[Mark] After a fruitless search for foul play,
3:06	if I was gonna steal round one from Nigel's acid,	if I was gonna steal round one from Nigel's acid,
3:08	it would come down to my wild-card element of destruction,	it would come down to my wild-card element of destruction,
3:12	which just so happened to be 50 piranhas I'd located in a New Jersey pet store	which just so happened to be 50 piranhas I'd located in a New Jersey pet store
3:16	that took less than a minute to clear a drumstick down to the bone.	that took less than a minute to clear a drumstick down to the bone.
3:20	My plan was ruined when I went to ship the piranhas	My plan was ruined when I went to ship the piranhas
3:23	because, apparently, in California, they're illegal to own as pets.	because, apparently, in California, they're illegal to own as pets.
3:27	Fortunately, the pet store informed me that bucktooth tetras, while smaller,	Fortunately, the pet store informed me that bucktooth tetras, while smaller,
3:31	with enough of 'em, would apparently do just as much damage.	with enough of 'em, would apparently do just as much damage.
3:34	-[William] Whoa! It's instant. -[Nigel] No way.	-[William] Whoa! It's instant. -[Nigel] No way.
3:37	[Mark] And while there was a nibble frenzy right when I put it in,	[Mark] And while there was a nibble frenzy right when I put it in,
3:40	when I raised it up, it was basically untouched,	when I raised it up, it was basically untouched,
3:43	which was embarrassing not only for me, but frankly for California.	which was embarrassing not only for me, but frankly for California.
3:46	You said at the beginning you'd eat whatever's left.	You said at the beginning you'd eat whatever's left.
3:49	-I never said that. -[laughter]	-I never said that. -[laughter]
3:51	[Mark] Which gave acid the early lead, sending us into round two with the iPad.	[Mark] Which gave acid the early lead, sending us into round two with the iPad.
3:55	Kevin and his lava were hungry for a W, and that hunger only intensified	Kevin and his lava were hungry for a W, and that hunger only intensified

3:59	when they saw the C-tier YouTube content playing on the iPad.	when they saw the C-tier YouTube content playing on the iPad.
4:02	-Right on my face. -Sorry, Will.	-Right on my face. -Sorry, Will.
4:04	[Mark] Oh no!	[Mark] Oh no!
4:12	-Oh! -[laughs]	-Oh! -[laughs]
4:13	-[Mark] Now he can't see if it's playing. - Oh, it's so hot!	-[Mark] Now he can't see if it's playing. - Oh, it's so hot!
4:16	-[Mark] Okay, that's enough. -No. More, more, more!	-[Mark] Okay, that's enough. -No. More, more, more!
4:20	[Mark] The molten rocks had melted the iPad, but for the real question,	[Mark] The molten rocks had melted the iPad, but for the real question,
4:23	was it enough to stop the iPad from playing?	was it enough to stop the iPad from playing?
4:27	Yes. Of course. That's going to do it.	Yes. Of course. That's going to do it.
4:29	Nigel was up next,	Nigel was up next,
4:31	and with more video content suitable for destruction loaded up,	and with more video content suitable for destruction loaded up,
4:34	he slipped the iPad into eight liters of concentrated acid.	he slipped the iPad into eight liters of concentrated acid.
4:40	[Mark] It was time to see what acid could do.	[Mark] It was time to see what acid could do.
4:45	-[Nigel] Oh man. -Boo!	-[Nigel] Oh man. -Boo!
4:48	-[Nigel] Hey, give it time, give it time! - Boo!	-[Nigel] Hey, give it time, give it time! - Boo!
4:50	[Mark] Even with a full hour soaking, not much happened	[Mark] Even with a full hour soaking, not much happened
4:53	besides some kind of small reaction that turned the acid some cool colors,	besides some kind of small reaction that turned the acid some cool colors,
4:57	which makes this the perfect opportunity to debunk the idea	which makes this the perfect opportunity to debunk the idea
5:00	that there are acids that can immediately dissolve all metals.	that there are acids that can immediately dissolve all metals.
5:04	You might say, "Hold up, Mark."	You might say, "Hold up, Mark."
5:05	"I've seen really popular videos like this online	"I've seen really popular videos like this online
5:08	where the metal spoon is immediately dissolved in acid."	where the metal spoon is immediately dissolved in acid."
5:11	However, the reason he could safely dip his fingers in the acid later in the video	However, the reason he could safely dip his fingers in the acid later in the video
5:16	is because that's not acid.	is because that's not acid.
5:18	It's watered-down Mountain Dew, and that spoon is made from gallium,	It's watered-down Mountain Dew, and that spoon is made from gallium,
5:21	which simply melts in lukewarm liquid.	which simply melts in lukewarm liquid.
5:23	Certain acids can dissolve certain metals,	Certain acids can dissolve certain metals,

5:26	but it just takes several hours for chunks of metal like this.	but it just takes several hours for chunks of metal like this.
5:30	With Nigel out of the running, it was time for the wild card.	With Nigel out of the running, it was time for the wild card.
5:33	Normally, this fires tennis balls. Today, we're firing golf balls	Normally, this fires tennis balls. Today, we're firing golf balls
5:37	at, like, 200 miles an hour.	at, like, 200 miles an hour.
5:39	I could catch it.	I could catch it.
5:40	I feel good on my destructive capabilities.	I feel good on my destructive capabilities.
5:42	The real question is, how's my aim?	The real question is, how's my aim?
5:47	Pretty good as it turns out.	Pretty good as it turns out.
5:51	-[laughs] -[Nigel] Oh my...	-[laughs] -[Nigel] Oh my...
5:52	Yeah!	Yeah!
5:55	Even after all this, it still had enough kinetic energy	Even after all this, it still had enough kinetic energy
5:58	to dent the back wall.	to dent the back wall.
5:59	-Wow! -It almost split the whole thing in half.	-Wow! -It almost split the whole thing in half.
6:02	This looks very violent. Let's see your acid do that, Nigel.	This looks very violent. Let's see your acid do that, Nigel.
6:06	-Is this a W? -It's pretty cool.	-Is this a W? -It's pretty cool.
6:08	Okay.	Okay.
6:10	[Mark] This meant I was tied for first as we headed into the ice round.	[Mark] This meant I was tied for first as we headed into the ice round.
6:14	-[Kevin] Are you ready? -Fire versus ice!	-[Kevin] Are you ready? -Fire versus ice!
6:17	And lava came right out of the gate, sizzling hot.	And lava came right out of the gate, sizzling hot.
6:19	-[Kevin] Oh, wait. -[laughter]	-[Kevin] Oh, wait. -[laughter]
6:21	[Kevin] No! No!	[Kevin] No! No!
6:23	-[Mark] Weak! Weak! -[Kevin] No!	-[Mark] Weak! Weak! -[Kevin] No!
6:25	-He didn't make it through one! -[Kevin] Look at it bubble though!	-He didn't make it through one! -[Kevin] Look at it bubble though!
6:29	[Mark] The bubbles were pretty cool,	[Mark] The bubbles were pretty cool,
6:31	but other than that, Kevin was grasping at straws.	but other than that, Kevin was grasping at straws.
6:33	[Kevin] I made lava spaghetti.	[Kevin] I made lava spaghetti.
6:35	[Mark] The ref tallied the final count as only breaking one of the 12 layers...	[Mark] The ref tallied the final count as only breaking one of the 12 layers...
6:42	[Mark] ...noting that the final break was too little, too late,	[Mark] ...noting that the final break was too little, too late,
6:45	which made it acid's turn.	which made it acid's turn.
6:47	Almost immediately...	Almost immediately...
6:48	Oh wow!	Oh wow!
6:49	[Mark] ...it tied lava.	[Mark] ...it tied lava.

6:50	-Oh, it's through! It's through. -It's through! Wow!	-Oh, it's through! It's through. -It's through! Wow!
6:54	I'm trying to aim.	I'm trying to aim.
6:56	[Mark] It just kept breaking through each layer	[Mark] It just kept breaking through each layer
6:58	as the acid mixed with the water,	as the acid mixed with the water,
7:00	creating an exothermic reaction that bored a hole like a laser.	creating an exothermic reaction that bored a hole like a laser.
7:04	-One more! One more! -One more! Why am I rooting for you?	-One more! One more! -One more! Why am I rooting for you?
7:08	[Mark] Ultimately, Nigel's acid made it through every last sheet of ice,	[Mark] Ultimately, Nigel's acid made it through every last sheet of ice,
7:11	which meant my wild card would have to do the same and do it faster.	which meant my wild card would have to do the same and do it faster.
7:15	Gentlemen, my approach here is a little bit more Kevin than Nigel.	Gentlemen, my approach here is a little bit more Kevin than Nigel.
7:20	-Whoa! -[Mark laughs]	-Whoa! -[Mark laughs]
7:27	-I feel like that did nothing. -You're making water.	-I feel like that did nothing. -You're making water.
7:30	All right. A good engineer always has a backup plan.	All right. A good engineer always has a backup plan.
7:34	[Kevin] Oh!	[Kevin] Oh!
7:36	[Mark] As a delivery of potential energy from the Acme Corporation...	[Mark] As a delivery of potential energy from the Acme Corporation...
7:40	[Kevin] Oh!	[Kevin] Oh!
7:42	[Mark] ...not only finished off my ice	[Mark] ...not only finished off my ice
7:45	but also took care of Nigel's.	but also took care of Nigel's.
7:46	-[Nigel] He killed mine too. -Technical foul.	-[Nigel] He killed mine too. -Technical foul.
7:49	-No, that was a regular foul. -What?	-No, that was a regular foul. -What?
7:51	But with further evidence of referee bias...	But with further evidence of referee bias...
7:53	-Flamethrower. Not flamethrower and anvil. -You're ruining everything.	-Flamethrower. Not flamethrower and anvil. -You're ruining everything.
7:57	[Mark] ...the win went to Nigel, which put acid in first place,	[Mark] ...the win went to Nigel, which put acid in first place,
8:00	headed to the toy city.	headed to the toy city.
8:02	And this one for sure had Toy Story vibes going on...	And this one for sure had Toy Story vibes going on...
8:05	[Kevin] Three, two, one.	[Kevin] Three, two, one.
8:06	[Mark] Whoa!	[Mark] Whoa!
8:08	...which definitely makes Kevin Backyard Sid.	...which definitely makes Kevin Backyard Sid.
8:11	-[Kevin] Whoa, it's going for the school! -[Mark] Oh no! For the school.	-[Kevin] Whoa, it's going for the school! -[Mark] Oh no! For the school.
8:16	Whoa! Whoa!	Whoa! Whoa!
8:18	-[Kevin] Oh no, Stretch! -[Mark] Stretch!	-[Kevin] Oh no, Stretch! -[Mark] Stretch!

8:20	I eventually swooped in to save the townspeople.	I eventually swooped in to save the townspeople.
8:23	I got it!	I got it!
8:27	I'm a hero!	I'm a hero!
8:28	But arguably it was...	But arguably it was...
8:29	-[Kevin] Oh, Stretch! -[Mark] Ah.	-[Kevin] Oh, Stretch! -[Mark] Ah.
8:31	...too little, too late. -[William] I don't think he'll make it.	...too little, too late. -[William] I don't think he'll make it.
8:35	[Mark] With an identical set of toys ready,	[Mark] With an identical set of toys ready,
8:37	it was more bad news for the action figures...	it was more bad news for the action figures...
8:39	-[Nigel] Start with sulfuric. -[Mark] Sulfuric?	-[Nigel] Start with sulfuric. -[Mark] Sulfuric?
8:41	...as the forecast called for acid rain.	...as the forecast called for acid rain.
8:44	-[Nigel] Oh! What? -[Kevin] Okay.	-[Nigel] Oh! What? -[Kevin] Okay.
8:46	He has red hair now.	He has red hair now.
8:47	[Mark] After a combination of several acids, the townspeople	[Mark] After a combination of several acids, the townspeople
8:51	were mildly inconvenienced at best.	were mildly inconvenienced at best.
8:55	[Kevin] Stretch's hair is back to yellow.	[Kevin] Stretch's hair is back to yellow.
8:57	[Mark] The lack of reaction with the plastic toys makes sense in hindsight.	[Mark] The lack of reaction with the plastic toys makes sense in hindsight.
9:01	Polymers have stable chemical structures, so acid can't break them down,	Polymers have stable chemical structures, so acid can't break them down,
9:05	which is not only the reason it takes them so long to degrade in a landfill,	which is not only the reason it takes them so long to degrade in a landfill,
9:08	but it's also why most acids are, therefore, stored in plastic containers.	but it's also why most acids are, therefore, stored in plastic containers.
9:13	In contrast to that lame result, it was time for the Mark Rober wild card...	In contrast to that lame result, it was time for the Mark Rober wild card...
9:17	-[chuckling] -Yeah.	-[chuckling] -Yeah.
9:19	[Mark] ...in the form of a two-ton steamroller.	[Mark] ...in the form of a two-ton steamroller.
9:33	-[William] Yeah! -[Kevin] Yes!	-[William] Yeah! -[Kevin] Yes!
9:35	[Mark] This makes a great time to mention	[Mark] This makes a great time to mention
9:38	that if you haven't been to CrunchLabs yet,	that if you haven't been to CrunchLabs yet,
9:41	you can now visit CrunchLabs for free in Roblox, but it's way cooler than that.	you can now visit CrunchLabs for free in Roblox, but it's way cooler than that.
9:45	The game's called Build a Rocket VS Mark Rober,	The game's called Build a Rocket VS Mark Rober,
9:48	where you build fully customizable rockets with your friends	where you build fully customizable rockets with your friends
9:51	and use your engineering skills	and use your engineering skills

9:52	to help me defeat the evil Fat Tail and rescue my squirrel friends.	to help me defeat the evil Fat Tail and rescue my squirrel friends.
9:56	We've been secretly working on this for over a year,	We've been secretly working on this for over a year,
9:59	and I know people are gonna get mad	and I know people are gonna get mad
10:01	and say I'm just doing this as a trick to teach you science.	and say I'm just doing this as a trick to teach you science.
10:04	And to that I say, "How dare you?"	And to that I say, "How dare you?"
10:06	I'm just trying to make a really fun free game,	I'm just trying to make a really fun free game,
10:08	and if you happen to learn about projectile motion and Newton's second law,	and if you happen to learn about projectile motion and Newton's second law,
10:12	that's not my fault.	that's not my fault.
10:14	And anyone who passes the game with the help of Phat Gus	And anyone who passes the game with the help of Phat Gus
10:17	before the end of the year can enter for a chance to win a ticket	before the end of the year can enter for a chance to win a ticket
10:20	to come out to CrunchLabs in real life for a day.	to come out to CrunchLabs in real life for a day.
10:23	We'll make a real-life version of the rocket you design.	We'll make a real-life version of the rocket you design.
10:26	So visit CrunchLabs.com/Roblox with a parent to learn how you can enter,	So visit CrunchLabs.com/Roblox with a parent to learn how you can enter,
10:30	then go play Build a Rocket on Roblox for free	then go play Build a Rocket on Roblox for free
10:32	so you can learn stuff.	so you can learn stuff.
10:34	I mean, just have fun building rockets with your friends.	I mean, just have fun building rockets with your friends.
10:37	Now, where were we?	Now, where were we?
10:38	-[laughter] -Oh no!	-[laughter] -Oh no!
10:43	No!	No!
10:45	This is very unsportsmanlike.	This is very unsportsmanlike.
10:52	-[William] No! -[Nigel] Oh no!	-[William] No! -[Nigel] Oh no!
10:54	Yeah!	Yeah!
10:57	I basically just pushed the stuff around, but it still felt cool.	I basically just pushed the stuff around, but it still felt cool.
11:01	[William] One Mark Rober versus three Buzz Lightyears.	[William] One Mark Rober versus three Buzz Lightyears.
11:04	Who'd have thought the Buzz Lightyears would win?	Who'd have thought the Buzz Lightyears would win?
11:06	This goes to Kevin.	This goes to Kevin.
11:08	[Mark] This is the bat challenge. Choose wood or metal.	[Mark] This is the bat challenge. Choose wood or metal.
11:11	Definitely, I think, metal. [chuckles]	Definitely, I think, metal. [chuckles]
11:14	-[Kevin] Oh, wow. -That made a sound.	-[Kevin] Oh, wow. -That made a sound.

11:16	[Mark] Kevin's lava quickly melted away the top half of the bat,	[Mark] Kevin's lava quickly melted away the top half of the bat,
11:19	as he topped things off...	as he topped things off...
11:21	That's a nice layer, Kevin.	That's a nice layer, Kevin.
11:22	...with a gratuitous flourish.	...with a gratuitous flourish.
11:24	So it was up to Nigel's acid to top that.	So it was up to Nigel's acid to top that.
11:27	[William] This time, you got it, Nigel.	[William] This time, you got it, Nigel.
11:31	[Mark] Yeah.	[Mark] Yeah.
11:32	At this point, it's me, and I'm gonna go with the wood bat	At this point, it's me, and I'm gonna go with the wood bat
11:35	'cause my contraption just happens to be	'cause my contraption just happens to be
11:39	a 2,000-watt laser.	a 2,000-watt laser.
11:42	Which you may recognize 'cause this is Kevin's laser.	Which you may recognize 'cause this is Kevin's laser.
11:45	It's true, and it's also how he knew a fact that I didn't.	It's true, and it's also how he knew a fact that I didn't.
11:49	You shouldn't have chose wood.	You shouldn't have chose wood.
11:51	[Mark] This 2,000-watt laser is actually terrifyingly effective	[Mark] This 2,000-watt laser is actually terrifyingly effective
11:55	against metallic objects.	against metallic objects.
11:56	So we ran a test.	So we ran a test.
12:01	Wow!	Wow!
12:02	Clearly, I need to swap for the metal bat.	Clearly, I need to swap for the metal bat.
12:05	Judge, can I--	Judge, can I--
12:06	-No. -[Mark] Now you see why he wasn't invited.	-No. -[Mark] Now you see why he wasn't invited.
12:09	Regardless, I pressed onward.	Regardless, I pressed onward.
12:11	All right.	All right.
12:13	Oh my goodness.	Oh my goodness.
12:14	That's pretty strong. Here I go.	That's pretty strong. Here I go.
12:16	And this was honestly the closest thing to a real-life lightsaber	And this was honestly the closest thing to a real-life lightsaber
12:20	I've ever experienced.	I've ever experienced.
12:22	Oh my God, he's two-thirds through, I think.	Oh my God, he's two-thirds through, I think.
12:26	[William] Oh!	[William] Oh!
12:27	Heck yeah, baby!	Heck yeah, baby!
12:30	Look at that. Pure destruction.	Look at that. Pure destruction.
12:32	I think that that is really cool, and Kevin's laser did it.	I think that that is really cool, and Kevin's laser did it.
12:37	[Mark] With that questionable ruling, if I didn't win the diamond round,	[Mark] With that questionable ruling, if I didn't win the diamond round,

12:40	I couldn't outright win this competition, even with the finale worth two points.	I couldn't outright win this competition, even with the finale worth two points.
12:45	1.2 carats right here.	1.2 carats right here.
12:47	-I do? -I don't.	-I do? -I don't.
12:48	-[Mark] Kevin kicked things off... -[Kevin] Ready?	-[Mark] Kevin kicked things off... -[Kevin] Ready?
12:51	[Mark] ...as we completely encased the diamond in sort of a lava sandwich.	[Mark] ...as we completely encased the diamond in sort of a lava sandwich.
12:55	Wow.	Wow.
12:56	And after it cooled off and we chiseled away the actual obsidian...	And after it cooled off and we chiseled away the actual obsidian...
13:00	Nothing for lava.	Nothing for lava.
13:01	...the diamond was perfectly intact.	...the diamond was perfectly intact.
13:03	1.2-carat diamond versus acid.	1.2-carat diamond versus acid.
13:09	-[music stops] -Yeah.	-[music stops] -Yeah.
13:10	-It's working. -[both laugh]	-It's working. -[both laugh]
13:12	[Mark] Classic acid.	[Mark] Classic acid.
13:13	Time for the wild card to kick it old school	Time for the wild card to kick it old school
13:17	and get back to basics.	and get back to basics.
13:19	I'm gonna smash the crud outta this. Just to reclaim my honor with the W,	I'm gonna smash the crud outta this. Just to reclaim my honor with the W,
13:23	heading into the final challenge,	heading into the final challenge,
13:25	in three, two, one.	in three, two, one.
13:30	-I missed. -[laughter]	-I missed. -[laughter]
13:33	-Did you miss again, Mark? -No!	-Did you miss again, Mark? -No!
13:36	Are you kidding me?	Are you kidding me?
13:38	That round's a wash.	That round's a wash.
13:39	You have two points. You have two points.	You have two points. You have two points.
13:42	You have no points 'cause nobody invited you.	You have no points 'cause nobody invited you.
13:45	And I've got one.	And I've got one.
13:46	Which set up the perfect finale showdown	Which set up the perfect finale showdown
13:49	where we each brought our own junkyard car.	where we each brought our own junkyard car.
13:53	-Where's Mark? -Where is Mark?	-Where's Mark? -Where is Mark?
14:00	That's probably Mark.	That's probably Mark.
14:05	Yee-ha-ha-ha-ha!	Yee-ha-ha-ha-ha!
14:08	-Yeah! -[Kevin] Oh!	-Yeah! -[Kevin] Oh!
14:09	He hit my car!	He hit my car!
14:11	Dude, why'd you park so close? [laughs]	Dude, why'd you park so close? [laughs]
14:13	You only know how to drive a Mars rover.	You only know how to drive a Mars rover.
14:15	[Mark] The rules were simple.	[Mark] The rules were simple.

14:17	Whoever could disable their car fastest would get two points and win it all.	Whoever could disable their car fastest would get two points and win it all.
14:21	Starting with Nigel, who was pulling out all the stops.	Starting with Nigel, who was pulling out all the stops.
14:24	I have a propane burner here	I have a propane burner here
14:26	with a nice big beaker loaded with hydrochloric acid.	with a nice big beaker loaded with hydrochloric acid.
14:29	I'm gonna start adding nitric acid.	I'm gonna start adding nitric acid.
14:31	This will make the same mixture that can dissolve gold. Aqua regia.	This will make the same mixture that can dissolve gold. Aqua regia.
14:34	Start the engine. We'll get this aqua regia nice and hot.	Start the engine. We'll get this aqua regia nice and hot.
14:38	It'll start tearing all the metal apart. To trigger it, I have something special.	It'll start tearing all the metal apart. To trigger it, I have something special.
14:42	[Mark] To pour the acid from a safe distance, he had a clever solution.	[Mark] To pour the acid from a safe distance, he had a clever solution.
14:46	But before he could get to that, we were rudely interrupted.	But before he could get to that, we were rudely interrupted.
14:49	Wow, more uninvited science YouTubers. Estefannie and Allen.	Wow, more uninvited science YouTubers. Estefannie and Allen.
14:54	Since they were already here, I put 'em to work	Since they were already here, I put 'em to work
14:57	with more questionably-sized referee uniforms.	with more questionably-sized referee uniforms.
14:59	After which, Nigel was up.	After which, Nigel was up.
15:02	When acid comes in contact with the metal,	When acid comes in contact with the metal,
15:04	it rips away the metal atoms like a magnet.	it rips away the metal atoms like a magnet.
15:07	We call this corrosion.	We call this corrosion.
15:08	You can increase the corrosive destructive power	You can increase the corrosive destructive power
15:10	by either, one, increasing the concentration of the acid,	by either, one, increasing the concentration of the acid,
15:13	meaning more magnets,	meaning more magnets,
15:14	or two, you heat it up.	or two, you heat it up.
15:15	If a liquid's hot, the molecules are moving around way more,	If a liquid's hot, the molecules are moving around way more,
15:19	so you increase the chance an acid molecule	so you increase the chance an acid molecule
15:21	will make contact with the metal and attract away an atom.	will make contact with the metal and attract away an atom.
15:23	In this case, as a last-ditch effort, Nigel did both.	In this case, as a last-ditch effort, Nigel did both.
15:27	Now he just had to shatter the beaker.	Now he just had to shatter the beaker.
15:37	-Nigel! -This was your plan?	-Nigel! -This was your plan?

15:39	Why does it just make holes?	Why does it just make holes?
15:41	-[Nigel] I like the slow release. -[Mark] There it goes!	-[Nigel] I like the slow release. -[Mark] There it goes!
15:46	-That is terrifying. -That is...	-That is terrifying. -That is...
15:48	[Mark] I've never seen a gas that color before.	[Mark] I've never seen a gas that color before.
15:50	Now for the verdict.	Now for the verdict.
15:52	It... It's still running.	It... It's still running.
15:56	Honda makes a good car.	Honda makes a good car.
15:58	And as Nigel released his frustrations...	And as Nigel released his frustrations...
16:03	-Oh! -[William] Do you feel better, Nigel?	-Oh! -[William] Do you feel better, Nigel?
16:06	[Mark] ...I prepared to release the final mega wild card	[Mark] ...I prepared to release the final mega wild card
16:09	in the form of a 50-ton excavator.	in the form of a 50-ton excavator.
16:13	And as I figure out the controls...	And as I figure out the controls...
16:15	Oh!	Oh!
16:16	I got it.	I got it.
16:17	...I want to remind the viewer	...I want to remind the viewer
16:19	the purpose of this video was to use tools of scientific inquiry	the purpose of this video was to use tools of scientific inquiry
16:22	for the noble cause	for the noble cause
16:23	of furthering our understanding of interactions in the natural world.	of furthering our understanding of interactions in the natural world.
16:27	Not just to live out my lifelong dream of operating a 50-ton excavator.	Not just to live out my lifelong dream of operating a 50-ton excavator.
16:32	-Arm tank! Arm tank! Arm tank! -[exclaims]	-Arm tank! Arm tank! Arm tank! -[exclaims]
16:35	-[Mark laughs] -Oh!	-[Mark laughs] -Oh!
16:38	[Mark] Oh, yeah! Stay down! Yes!	[Mark] Oh, yeah! Stay down! Yes!
16:45	-Oh no! -[car horn honks]	-Oh no! -[car horn honks]
16:50	Put it out of its misery, Mark.	Put it out of its misery, Mark.
16:53	[Mark] Yeah!	[Mark] Yeah!
16:55	Woo!	Woo!
16:57	Yeah! [laughs]	Yeah! [laughs]
17:01	-[all] Yeah! -Oh!	-[all] Yeah! -Oh!
17:03	[Mark] And with my finishing move...	[Mark] And with my finishing move...
17:05	-[Allen] Ooh, there you go! -That was pretty good.	-[Allen] Ooh, there you go! -That was pretty good.
17:08	I am very precariously balanced.	I am very precariously balanced.
17:12	...while the horn was still working, the engine was definitely not,	...while the horn was still working, the engine was definitely not,
17:16	and I returned to receive the praise from my contemporaries.	and I returned to receive the praise from my contemporaries.
17:19	-The machine was impressive. -[laughter]	-The machine was impressive. -[laughter]

17:22	-All right. -Lava.	-All right. -Lava.
17:24	[Mark] So for the grand finale, lava.	[Mark] So for the grand finale, lava.
17:29	[all] Oh!	[all] Oh!
17:40	[Mark] With the lava poured, time for Kevin's escape.	[Mark] With the lava poured, time for Kevin's escape.
17:46	[Kevin] Agh!	[Kevin] Agh!
17:48	[Mark] Oh, it's stopped already. No!	[Mark] Oh, it's stopped already. No!
17:51	-[Kevin] As soon as I poured it on. -[Mark] Serious?	-[Kevin] As soon as I poured it on. -[Mark] Serious?
17:53	-All it took was four gallons of lava. - [laughter]	-All it took was four gallons of lava. - [laughter]
17:57	Oh, something's happening!	Oh, something's happening!
17:59	Oh wow!	Oh wow!
18:01	Luckily, we had fire extinguishers on hand...	Luckily, we had fire extinguishers on hand...
18:04	-[music stops] -...that basically did nothing.	-[music stops] -...that basically did nothing.
18:07	-[laughs] -[high-pitched tone]	-[laughs] -[high-pitched tone]
18:08	[Mark] The judges just had to confer amongst themselves...	[Mark] The judges just had to confer amongst themselves...
18:11	Don't make me regret having invited any of you in the first place.	Don't make me regret having invited any of you in the first place.
18:14	-...and deliver the verdict. -Okay, so...	-...and deliver the verdict. -Okay, so...
18:16	-Kevin, your car stopped in four seconds. -Nigel's car still runs perfectly.	-Kevin, your car stopped in four seconds. -Nigel's car still runs perfectly.
18:21	-Still running. -It could stop at any moment.	-Still running. -It could stop at any moment.
18:23	-And Mark? -How long?	-And Mark? -How long?
18:25	Like, ten minutes to even get on the thing.	Like, ten minutes to even get on the thing.
18:28	You counted from when I start moving?	You counted from when I start moving?
18:30	Which honestly feels like a technicality, but I'm nothing...	Which honestly feels like a technicality, but I'm nothing...
18:33	-Kevin! -Kevin!	-Kevin! -Kevin!
18:35	[Mark] ...if not a gracious loser.	[Mark] ...if not a gracious loser.
18:37	As a reward for a fight well fought,	As a reward for a fight well fought,
18:39	I gave Kevin the indestructible 1.2-carat diamond.	I gave Kevin the indestructible 1.2-carat diamond.
18:42	Congratulations.	Congratulations.
18:43	Now go drive off in your cars.	Now go drive off in your cars.
18:45	You're welcome. I'm basically Oprah.	You're welcome. I'm basically Oprah.
18:49	Given that we've been secretly working on this Roblox game for over a year,	Given that we've been secretly working on this Roblox game for over a year,
18:53	I'm super stoked to show you, in an exclusive world premiere,	I'm super stoked to show you, in an exclusive world premiere,

18:56	the official game trailer.	the official game trailer.
18:58	Build a Rocket. The sky's not the limit. It's just the beginning.	Build a Rocket. The sky's not the limit. It's just the beginning.
19:01	I'm Mark Rober, and my new game, Build a Rocket, is your key	I'm Mark Rober, and my new game, Build a Rocket, is your key
19:05	to a super cool space adventure.	to a super cool space adventure.
19:07	Build rockets at CrunchLabs, journey to distant planets,	Build rockets at CrunchLabs, journey to distant planets,
19:10	mine for valuable resources, then build even cooler mega-rockets	mine for valuable resources, then build even cooler mega-rockets
19:14	to explore the furthest corners of the universe.	to explore the furthest corners of the universe.
19:16	You can build with your friends,	You can build with your friends,
19:18	customize your rocket with unlimited combinations,	customize your rocket with unlimited combinations,
19:20	navigate treacherous space environments,	navigate treacherous space environments,
19:23	then use your mad engineering skills to defeat the evil rat Fat Tail	then use your mad engineering skills to defeat the evil rat Fat Tail
19:26	to get my squirrel friends back.	to get my squirrel friends back.
19:28	Building your first rocket's easy, but if you wanna master the game,	Building your first rocket's easy, but if you wanna master the game,
19:31	you need to think like an engineer	you need to think like an engineer
19:33	and balance your rocket's mass, thrusters, fuel supply, and navigation capabilities.	and balance your rocket's mass, thrusters, fuel supply, and navigation capabilities.
19:38	As you explore each planet, you'll find unique challenges and rewards.	As you explore each planet, you'll find unique challenges and rewards.
19:41	Will you dig deep into the alien soil of the Martian surface	Will you dig deep into the alien soil of the Martian surface
19:45	or make discoveries on uncharted planets	or make discoveries on uncharted planets
19:47	like Twistopia or Balloonanza or Marshmallia?	like Twistopia or Balloonanza or Marshmallia?
19:50	That's totally up to you.	That's totally up to you.
19:52	In between missions, hang out with your friends at CrunchLabs.	In between missions, hang out with your friends at CrunchLabs.
19:55	You might spot some Easter eggs,	You might spot some Easter eggs,
19:57	including hints at whatever my next video will be.	including hints at whatever my next video will be.
19:59	And CrunchLabs is a real place.	And CrunchLabs is a real place.
20:01	It's where I come up with the ideas and builds for all my videos.	It's where I come up with the ideas and builds for all my videos.
20:05	Anyone who beats the game this year has a chance to win a platinum ticket	Anyone who beats the game this year has a chance to win a platinum ticket
20:08	to come to CrunchLabs in real life for a day,	to come to CrunchLabs in real life for a day,

20:10	where we'll build an actual version of the rocket you designed in the game.	where we'll build an actual version of the rocket you designed in the game.
20:14	Visit CrunchLabs.com/Roblox with a parent to learn how to enter	Visit CrunchLabs.com/Roblox with a parent to learn how to enter
20:17	or you can just play Build a Rocket for free on Roblox.	or you can just play Build a Rocket for free on Roblox.
20:20	I'll be there playing myself a bunch,	I'll be there playing myself a bunch,
20:23	and I can't wait to see the super clever ideas	and I can't wait to see the super clever ideas
20:25	that you come up with. Thanks for watching.	that you come up with. Thanks for watching.